



Choosing multichannel objective response detectors for multichannel auditory steady-state responses

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ABSTRACT

The assessment of evoked responses still largely relies on subjective assessment, though a number of objective response detection techniques have been proposed, which are usually based on the statistical analysis of electroencephalogram (EEG) signals. In the current work, we expand on this with an investigation of multichannel objective response detection (MORD) methods, by assessing the performance of six previously published MORD algorithms together with three new methods (named product of Local Spectral F-test, average and the product of Component Synchrony Measure). The MORD techniques were evaluated in both simulated data and in multi-frequency Auditory Steady-State Responses (ASSRs) recorded from 11 normal-hearing volunteers. The performances of MORD techniques were evaluated based on detection rates and the number of false positive tests. The results demonstrate that inflated false positive rates can arise when correlation between EEG channels is not taken into account by some of the MORD methods, but the algorithms are fairly robust to low cross-channel correlations. Furthermore, two of the already established MORD techniques showed immunity to cross-channel correlation. The results showed that the sensitivity of response detection increased when more than one channel was used, and an increase of up to 16 % compared to the best single-channel method could be achieved by the newly proposed best MORD methods. MORD methods are thus a promising new tool for ASSR detection.

1. Introduction

In the assessment of hearing, evoked responses have gained increasing interest, as they do not require subjective (voluntary) responses from the listener, which makes them suitable for a range of patient groups that are unable or unwilling to cooperate in conventional behavioural tests. The Auditory Steady-State Response (ASSR) is an electrophysiological response evoked by periodic acoustic stimuli, and is characterized by spectral components that are stable in amplitude and phase over time [1–3], coupled to the stimulus. ASSRs evoked by Amplitude-Modulated (AM) and/or Frequency-Modulated (FM) tones present increased energy at the modulation frequency and its harmonics. In studies that investigated the influence of the modulation frequency on

ASSR amplitudes, 70–110 Hz (around 80 Hz) modulators were reported to be less affected by sleep [4,5] and are thus favoured for clinical use, especially in children [2,6]. ASSRs have the great advantage over transient evoked response methods (e.g. repeated tone-pips), in that multiple stimulus frequencies can be evaluated simultaneously (commonly called multifrequency ASSRs), which can significantly reduce the time required for hearing assessment [7].

In clinical applications, the ASSRs can be recorded by means of the electroencephalogram (EEG), and the presence or absence of response can be inferred by applying Objective Response Detection (ORD) techniques that allow an automatic, frequency-specific assessment of hearing thresholds [8–12]. ORD techniques are usually statistical methods, with the detection of a response being based on the rejection of the null

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hypothesis (H_0), that no response is present [13,14]. The presence of a response is thus deemed to occur when the value of some variable calculated from the signal exceeds the critical value (at a chosen significance level α) that is obtained from its probability distribution under H_0 .

Most ORD techniques use information from only one EEG channel, i. e., they are univariate. In order to improve the sensitivity of the detector, more channels can be added, since the response can be usually noted in a number of different channels. In study [15], the authors proposed to Multichannel Component Synchrony Measure (MCSM) based on the phase angle of N EEG channel, the results showed an increase in the detection rate (sensitivity) as the number of EEG channels.

Miranda de Sá et al. in [16] proposed to Multichannel Magnitude-Squared Coherence (MMSC), which is a measure of the linear correlation between N EEG channels and the stimulation signal. In studies [17,18] evaluating the performance of the MMSC in the detection of ASSR, Felix et al. in [17] and Eloi et al. in [18] reported an increase in the detection rate of 13.5 % and 18 %, respectively, in relation to the use of the technique with only one signal. Rocha et al. in [19] proposed the Multichannel Local F-test, which led to the consistently higher detection rates in the EEG during photic stimulation in comparison with using a single channel. Other studies [20–23] have also led to an increase in the detection rate of detector with the use of information from more than one EEG channel. In a different scenario, results presented in [24,25] showed the advantages of using more than one signal in the photoelectrochemical analysis.

The main research question of this work is critically evaluating the performance of a wider range of MORD methods (including three new ones proposed here) in the detection of multifrequency ASSRs, in order to recommend the best techniques for future applications in the analysis of evoked potentials embedded in the multichannel EEG.

The paper is organized as follows: Section 2 is a review of single-channel ORD methods. Section 3 describes the MORD techniques (established and new) and shows how critical values are estimated. The methods for generating synthetic data for multi-channel ASSR, and for acquiring signals from human volunteers are also described, together with the performance criteria for assessing and comparing the different methods. Details are provided of the previously published and new MORD methods. The performance evaluation of the MORD techniques are detailed in Section 4 and discussed in Section 5. Finally, Section 6 presents the conclusions.

2. Objective response detection (ORD)

The single-channel ORD methods on which the multichannel methods are based will be described first. We will assume that the EEG signal ($y[k]$) was recorded using a stimulus with a frequency of f_c (carrier frequency), modulated at a frequency f_0 . The response in the EEG will thus be observed primarily at f_0 – which will thus be the main focus of analysis.

In order to calculate the Component Synchrony Measure (CSM) [26] and Magnitude-Squared Coherence (MSC) [27], the discrete-time EEG signal $y[k]$ is first divided into M equal-length, non-overlapping windows. The length of the windows is chosen so as to contain an integer number of cycles of the modulation frequency f_0 , to reduce the effect of spectral leakage at the stimulus frequency (and its harmonics). The Discrete Fourier Transform (DFT) is then applied to each of the windowed segments, $y_i[k]$, where i refers to the temporal location of the window in the recording, to give $Y_i(f_0)$. MSC and CSM seek quantify the extent to which these DFT components ($Y_i(f_0)$) are synchronized (or correlated) with the stimulus; the MSC uses both the magnitude and phase of $Y_i(f_0)$, whereas CSM only uses the phase for this purpose. CSM and MSC are then found from:

$$CSM(f_0) = \left[\frac{1}{M} \sum_{i=1}^M \cos(\theta_i(f_0)) \right]^2 + \left[\frac{1}{M} \sum_{i=1}^M \sin(\theta_i(f_0)) \right]^2, \quad (1)$$

$$MSC(f_0) = \frac{\left| \sum_{i=1}^M Y_i(f_0) \right|^2}{M \sum_{i=1}^M |Y_i(f_0)|^2}, \quad (2)$$

where $\theta_i(f_0)$ is the phase angle of $Y_i(f_0)$ at the modulation frequency f_0 and M is the number of windows used.

The Local F-test (LFT) was described in [28] and seeks to quantify how much larger the amplitude of the DFT is at the stimulus frequency, compared to its nearest neighbours:

$$LFT(f_0) = \frac{|Y(f_0)|^2}{\frac{1}{L} \sum_{\substack{k=0-L/2 \\ k \neq 0}}^{0+L/2} |Y(f_k)|^2}, \quad (3)$$

where $Y(f_0)$ and $Y(f_k)$ are the DFT coefficients calculated from a single DFT applied to the whole recording, at frequency f_0 , and its L closest neighbouring frequencies, respectively. Local here refers to the use of the L frequencies adjacent to f_0 . Critical values for the detection of a response can be obtained as described in section 4.2.

3. Material and methods

3.1. Multivariate objective response detection (MORD)

MORD techniques are described in the following sections, and are grouped by the type of information used in their estimation:

- Magnitude information: MLFT, Additive Local Spectral F-test (aLFT) and Product of LFTs (pLFT);
- Phase information: MCSM, Average of CSMs (aCSM) and Product of CSMs (pCSM);
- Phase and Magnitude information: MMSC, Average of CSMs (aMSC) and Product of CSMs (pMSC).

The three new methods proposed in this work are pLFT, aCSM and pCSM and will be presented together with the six established methods.

3.1.1. MORD based only on magnitude information

The Multichannel Local F-test (MLFT) for N channels ($y_1[k], y_2[k], \dots, y_N[k]$) is estimated by the following expression [19], in which the numerator and denominator of the single channel LFT are expanded to be sums across all channels:

$$MLFT(f_0) = \frac{\sum_{j=1}^N |Y_j(f_0)|^2}{\sum_{j=1}^N \left[\frac{1}{L} \sum_{\substack{k=0-L/2 \\ k \neq 0}}^{0+L/2} |Y_j(f_k)|^2 \right]}, \quad (4)$$

where N is the number of channels, $Y_j(f_0)$ and $Y_j(f_k)$ are Fourier Transforms of the j -th channel ($y_j[k]$) at frequency f_0 , or its L closest neighbouring frequencies (f_k).

An alternative method for multichannel extension of the LFT is the Additive Local Spectral F-test (aLFT), which can be obtained from Eq. 4 in [21] as

$$aLFT(f_0) = \sum_{j=1}^N LFT_j(f_0) \quad (5)$$

Here the single-channel LFT values are simply added, such that aLFT becomes large when multiple channels have large LFTs. We now propose a new alternative way to combine the LFT across channels by multiplication:

$$pLFT(f_0) = \prod_{j=1}^N LFT_j(f_0) \quad (6)$$

It may be noted that aLFT is linked to the arithmetic mean of the LFTs, and pLFT to the geometric mean (both without normalization) from all individual channels.

3.1.2. MORD based only on phase information

In a similar manner, the Multichannel Component Synchrony Measure (MCSM) expands on the single channel version, by using the average phase (across all channels) in each time-window. Thus for N discrete-time channels ($y_1[k]$, $y_2[k]$, ..., $y_N[k]$), divided into M non-overlapping windows, the MCSM is given as [15]:

$$MCSM(f_0) = \left[\frac{1}{M} \sum_{i=1}^M \cos(\bar{\theta}_i(f_0)) \right]^2 + \left[\frac{1}{M} \sum_{i=1}^M \sin(\bar{\theta}_i(f_0)) \right]^2, \quad (7)$$

where

$$\bar{\theta}_i(f_0) = \begin{cases} \tan^{-1}(\bar{S}_i/\bar{C}_i) & \text{if } \bar{C}_i \geq 0 \\ \tan^{-1}(\bar{S}_i/\bar{C}_i) + \pi & \text{if } \bar{C}_i < 0 \end{cases} \quad (8)$$

with $\bar{C}_i = (1/N) \sum_{j=1}^N \cos\theta_{ij}(f_0)$, $\bar{S}_i = (1/N) \sum_{j=1}^N \sin\theta_{ij}(f_0)$ and $\theta_{ij}(f_0)$ is the phase angle at frequency f_0 of the Fourier Transform of the i -th window in the j -th channel and M is again the number of windows. In the particular case of $N = 1$, the MCSM is reduced to the CSM. In addition, two extensions to MORD techniques are now proposed based on the average (aCSM) and product (pCSM) of CSMs from the individual channels, as follows:

$$aCSM(f_0) = \frac{1}{N} \sum_{j=1}^N CSM_j(f_0) \quad (9)$$

$$pMSC(f_0) = \prod_{j=1}^N CSM_j(f_0) \quad (10)$$

where $CSM_j(f_0)$ is the CSM of the j -th channel at frequency f_0 .

3.1.3. MORD based on both phase and magnitude information

The MMSC is an extension of the single-channel MSC in a manner equivalent to moving from a bivariate correlation coefficient to multiple correlation, i.e., seeking to find the strength of the linear association of the EEG responses (linearly combined across all channels in each window) with the stimulus. For N discrete-time channels ($y_1[k]$, $y_2[k]$, ..., $y_N[k]$), the MMSC is obtained with the following equation [16]

$$MMSC(f_0) = \frac{\mathbf{V}^H \hat{\mathbf{S}}_{yy}^{-1} \mathbf{V}}{M} \quad (11)$$

where

$$\mathbf{V}^H = \left[\sum_{i=1}^M Y_{i1}(f_0) \quad \sum_{i=1}^M Y_{i2}(f_0) \dots \sum_{i=1}^M Y_{iN}(f_0) \right], \quad (12)$$

and H indicates the Hermitian matrix (i.e., the conjugate transpose) operator, $Y_{ij}(f_0)$ ($j = 1, 2, \dots, N$) indicates the i -th window Fourier Transform for the j -th channel and $\hat{\mathbf{S}}_{yy}$ is the estimate of the cross-spectral

matrix of the N channels at frequency f_0 . For the case of $N = 1$, the expression of MMSC is reduced to that for the single-channel MSC [16]. Critical values for MMSC have been derived from theory [15] and take any correlation between EEG channels into account. In [29], we extended the single-channel MSC by proposing two new detection techniques based on the average (aMSC) and on the product (pMSC) of MSCs from two different signals (see Eq. 7 and 8 from [24]). In this paper, we expand these further to include N single-channel MSC estimates, as follows:

$$aMSC(f_0) = \frac{1}{N} \sum_{j=1}^N MSC_j(f_0) \quad (13)$$

$$pMSC(f_0) = \prod_{j=1}^N MSC_j(f_0) \quad (14)$$

where $MSC_j(f_0)$ is the single-channel MSC of the j -th channel.

3.2. Estimation of critical values

The critical values give the statistical thresholds for detection. These values are derived from the detector statistical distribution under the null hypothesis i.e., in the absence of a response. Often these distributions are known from theory (with appropriate assumptions such as Gaussianity and no correlation between background noise terms or among successive data windows) and the critical values can be calculated. For instance, this is the case for the MMSC, MCSM, MLFT and for their univariate versions. An alternative method commonly used for obtaining critical values is Monte Carlo simulation [20,30], which can be applied in situations where the detector statistical distribution is either unknown or where simplifying assumptions may not fully apply (e.g. due to preprocessing of data [31]) and the statistics become hence intractable in mathematical derivations. In this work, Monte Carlo simulations were used in order to obtain the critical values for all the detectors.

The Monte Carlo simulation was performed with 1,000,000 sets of normally distributed white signals (i.e., serially uncorrelated in each signal) with zero mean and unitary variance and with duration equal to that obtained in the recordings. These signals do not have a response present and thus aim to represent the signals under H_0 . As described in section 3.4.3, the same preprocessing techniques followed by MORD or ORD were applied to these sets of simulated [31]. The critical values were estimated by finding the percentile of the estimated MORD values corresponding to the desired significance level α (5 % in this work).

3.3. Synthetic data

Since EEG channels taken from different locations on a participant's scalp are likely to have some correlation, the effect of correlated noise on the false-positive (FP) rate obtained by MORD techniques should be assessed. To this end, Monte Carlo simulations were adapted. The simulations were performed with 1,000,000 sets of N channels, each set of channels was generated according to the following expression:

$$y_j[k] = r \cdot c[k] + \sqrt{1-r^2} \cdot n_j[k] \quad (15)$$

where, j is the channel index ($1, 2, \dots, N$), and c and n_j are both Gaussian noises with zero mean and unitary variance, following the usual assumption for EEG signals [23,32,33]. The noise $c[k]$ is a common parcel to all channels, and $n_j[k]$ is the uncorrelated noise between channels ($E[n_p \cdot n_q] = 0, \forall p \neq q$); r is the desired correlation coefficient between channels. When $r = 0$, the channels are uncorrelated. The MORD techniques were then applied using the critical values obtained from independent samples (see section 4.2) and a significance level of 5%. The FP rate for each MORD technique was then determined as

outlined in 'Performance Measures' (section 3.4.4). This was repeated for a range of correlation values r between 0 and 1, in order to assess the excess FPs that may be obtained when the assumption of independence is violated.

3.4. EEG signals

The methods were then also evaluated by using ASSR signals recorded from normal-hearing volunteers in order to assess detection rate (DR) and FP rate of the MORD techniques in detecting responses.

3.4.1. Multifrequency ASSR stimulation

The second order exponentially-modulated (AM2) tones evoke larger responses than simple (linear) AM stimuli in both infants and adults [34]. The AM2 tones are obtained by multiplying a carrier frequency (f_c) and the modulation envelope, with frequency f_0 . For a maximum amplitude A and modulation depth of 1 (100 %), the AM2 signal, $s(t)$, is given as [34]:

$$s(t) = A \sin 2\pi f_c t \cdot \left\{ 1 + 2 \left[\left(\frac{1 + \sin 2\pi f_0 t}{2} \right)^2 - 0.5 \right] \right\}. \quad (16)$$

The stimulation was presented through E-A-RTone 5A (Aearo Technologies, USA), with four carrier frequencies simultaneously applied to both ears. The carrier frequencies were 0.5, 1, 2 and 4 kHz and they were modulated at frequencies of 81, 85, 89 and 93 Hz for the right ear and 83, 87, 91, 95 Hz for the left ear, respectively, and the four modulated tones were then added before delivery to each ear. Thus, eight AM2 tones were used simultaneously to evoke the ASSRs.

Each volunteer attended one data acquisition session with stimulation at 50, 40 and 30 dB sound pressure level (SPL), with durations of 5, 8, and 8 min respectively, in addition to one additional session without any acoustic stimulation (only spontaneous EEG activity was recorded) with a duration of 1 min. Sound levels were calibrated with a sound level meter (Brüel & Kjær model 2250 with coupler 2 cc DB 0138, DKK).

3.4.2. Data acquisition

Eleven adult volunteers (2 females and 9 males) aged between 20 and 35 years, with normal hearing (thresholds ≤ 20 dB HL at frequencies between 500 and 4000 Hz), were recruited for the present study. The study was approved by the Local Ethics Committee (CEP/UFV No. 2.105.334). The data recording was performed in a soundproof cabin, located at the Interdisciplinary Nucleus for Signal Analysis (NIAS) of the Federal University of Viçosa (UFV, Brazil), and the volunteers were asked to lie on a bed with eyes closed, and to relax or sleep.

The EEG data were recorded by 16 scalp electrodes with a nape reference and F_{pz} as ground. The channels recorded were Fz, F3, F4, F7, FCz, C3, C4, Pz, P3, P4, T3, T4, T5, T6 and Oz. The electrodes were connected to an analog front-end (RHA2216, Intan Technologies, USA) and digitized using a ADS127L01 (Texas Instruments, USA) analogue to digital converter board. The high-pass and low-pass frequencies for the EEG signals were 0.5 and 300 Hz, respectively. The sampling rate was 1000 Hz per channel for 7 volunteers and 1750 Hz per channel for 4 volunteers. After the recordings, the EEGs signals were divided into windows of one second duration (resulting in a spectral resolution of 1 Hz for all volunteers) and stored on a hard disk for offline analysis. The window length was chosen so that the modulation frequencies of the stimuli have exactly an integer number of cycles in each window in order to minimize spectral leakage [35].

3.4.3. Preprocessing and detection of ASSR

Raw EEG data were filtered offline by an eighth-order band-pass Butterworth filter with cutoff frequencies one Hz above and below the upper ($f_0 - 1$) and lower ($f_0 + 1$) stimulus modulation frequencies, respectively. Next, artefact rejection was carried out by rejecting windows if their maximum absolute amplitude exceeded 40 μV . Then ORD

and MORD techniques were applied using the critical values estimated as described in section 3.2 for a significance level of 5%. In the case of MLFT, aLFT and pLFT, the number of neighbouring frequencies was $L = 12$, since these techniques are applied to the whole recording, then the side band is given by $L/(2 \times \text{recording time})$, e.g. a 5-minute recording length results in a side band of 0.02 Hz.

3.4.4. Performance measures

For each detector, the DR and FP rate were estimated. In the case of MORD detectors, the set of electrodes used in the estimation was varied and all combinations of 1–16 channels were tested. The DR was determined by:

$$DR = \frac{ND}{NT}, \quad (17)$$

where ND is the number of significant responses at the modulation frequencies and NT is the total number of possible detections (11 volunteers $\times$ 8 modulation frequencies $\times$ 3 stimulus intensities = 264 tests). FP rate was also estimated using Eq. 17 (with ND now referring to FP detections) using the background EEG collected prior to each stimulation run.

Two criteria were then used to select the best combinations of EEG electrodes for each MORD method: the first was that the detector must guarantee a FP rate of 5% or less in with the specific combination of channels used. Given the finite number (88) of tests performed, a nominal 5% FP rate would be expected to lead to between 3 and 6 detections from 88 tests under H_0 . These bounds can be obtained from the Binomial distribution (90 % critical values) with 88 tests and a probability of success equal to 5%. Thus, MORD methods that give fewer than 6 FPs (≤ 6.8 %) are deemed to give an acceptable FP rate. For the second criterion, the electrode combinations of each detector that passed the first criterion were ranked according to their DR. The best combinations of EEG electrodes for each MORD method was the one that obtained the highest detection rate and passed the test of FP rate. Differences in DR between methods were tested for statistical significance using McNemar's test.

4. Results

4.1. Synthetic data

The FP rate of the MORD techniques as a function of the between-channel correlation coefficient (r), with $\alpha = 5\%$ and the number of windows $M = 30$ (for MORDs that do not use windowing, a single window was used with the duration equivalent to the 30 concatenated windows), is shown in Fig. 1. It is clear that only MMSC and MCSM do not give an inflated FP rate, when between-channel correlations increase beyond about 0.5.

4.2. EEG signals

First, the FP rate was tested by using all possible combinations of channels, for a given total number of channels (from 1–16). Fig. 2 shows the percentage of combinations of channels used in MORD estimation that showed an acceptable FP rate (corresponding to a nominal value of 5%). Clearly, MCSM shows consistently the expected FP rate, followed closely by MMSC, but for other methods only a few channel combinations stay within the expected bounds of FPs. These results agree with those observed in the simulations, where only MMSC and MCSM were found to consistently give acceptable false positive rates. In the analysis of the second criterion (highest DR rate), only the electrode combinations of each detector that passed the first criterion were used. Fig. 3 shows the highest DR rate obtained with the MORD detectors as a function of N. For the univariate case, methods, the MSC (equivalent to MMSC, $N = 1$) showed the highest DR compared to CSM and LFT,

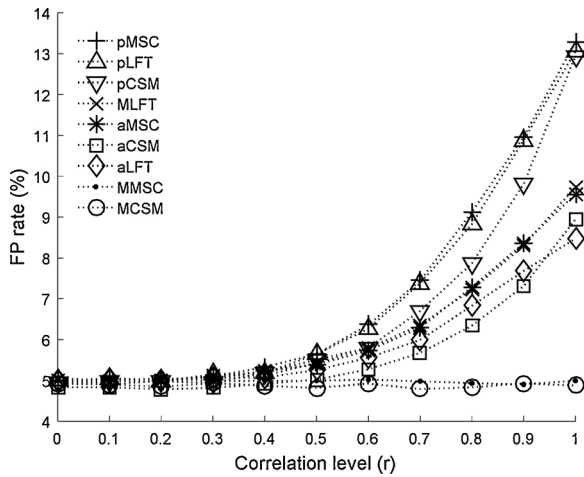


Fig. 1. False Positive (FP) rate of the MORD techniques as functions of correlation level (r). In this simulation, the number of windows was $M = 30$, the number of channels was $N = 2$ and the significance level (α) was set to 5%. Note that these results are based on 1,000,000 simulations, and only a small excess positive rate (FP rate $< 6\%$) is observed for between-channel correlation levels (see Eq. 15) below 0.5. MMSC and MCSM are both immune to correlations between channels.

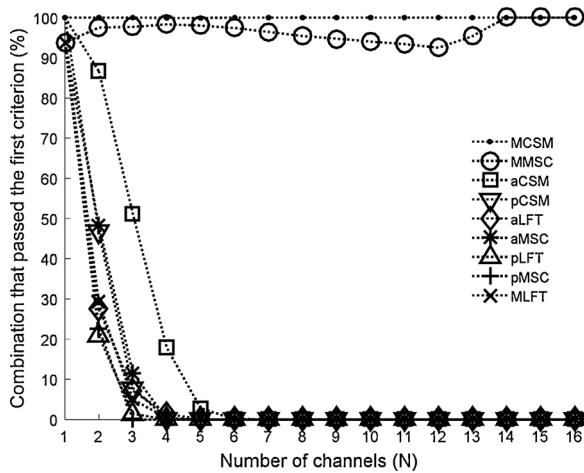


Fig. 2. Percentage of combinations of channels that kept the FP rate lower than the desired significance level, as a function of the number of EEG signals (N) for all nine MORD techniques.

against which the multivariate methods can be compared.

Table 1 shows the combinations of electrodes with the best performance (i.e., passed the test of FP rate and obtained the highest detection rate) for MORD techniques. For comparison purposes, the results from the best univariate case was included as the first line of the table. McNemar Test ($p = 0.05$ and 0.01) was used to test if there was statistically significant difference between the DR of the best univariate case and the MORD results. Fig. 3 suggests that aMSC achieves the highest detection rate with 3 EEG channels, but its DR is not significantly different to that of the best performance observed with aMSC ($N = 3$), pMSC ($N = 2$), aCSM ($N = 3$), MMSC ($N = 7$), aLFT ($N = 4$) and pCSM ($N = 3$).

5. Discussion

As is demonstrated in Appendix A.1, the MMSC estimates remain unchanged when this assumption is not justified, i.e. the MMSC estimates are unchanged by correlation between channels under H_0 . This

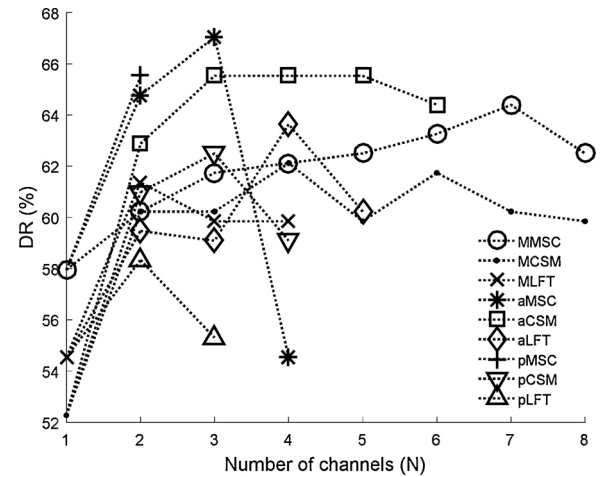


Fig. 3. The highest detection rate (DR) for each MORD technique and increasing number of EEG signals (N) with $\alpha = 0.05$. Only those channel combinations that had an acceptable false positive (FP) rate were tested. The absence of a marker indicates that no combinations of electrodes kept FP lower than 5%. The results for $N > 8$ are not shown to ease visualization: the DR for MMSC and MCSM did not rise again.

Table 1

Best combinations of EEG electrodes according to the performance criteria. The first line shows the best result for single channel techniques ($N = 1$). The third and fourth column show the Detection Rate (DR) and False Positive (FP) rate, respectively.

Techniques	Electrodes	DR(%)	FP rate(%)
MSC (univariate)	FC _z	57.9	4.3
aMSC	FC _z -P4-Oz	67.0**	6.3
pMSC	FC _z -P4	65.5**	6.7
aCSM	F4-Cz-Pz	65.5**	6.7
MMSC	FC _z -F4- Fz-T6-T5-P4-Pz	64.4*	3.8
aLFT	F4-F3-P4-Oz	63.6*	6.5
pCSM	F4-P4-Pz	62.5*	6.1
MCSM	FC _z -F4-Fz-Cz	62.1*	3.4
MLFT	FC _z -P4	61.6	6.1
pLFT	F4-P4	58.3	6.3

* McNemar Test ($p < 0.05$) show statistical difference in relation to MSC.

** McNemar Test ($p < 0.01$) show statistical difference in relation to MSC.

same propertie holds for MCSM under H_0 , i.e. the average phase angle is uniformly distributed in spite of the correlation between the channels for the case when of no periodic component in the data. See Appendix A.2). These, however, cannot be guaranteed for the remaining MORD.

In this work, the critical values of the MORD techniques were estimated using standard Monte Carlo simulations under the assumption of uncorrelated EEG activity in the different channels. As is demonstrated in Appendix A.1, the MMSC estimates remain unchanged when this assumption is not justified, i.e., MMSC estimates are unchanged by correlation between channels. This same property holds for MCSM under H_0 , i.e., the average phase angle is uniformly distributed in spite of the correlation between the channels for the case when of no periodic component in the data (see Appendix A.2). These, however, cannot be guaranteed for the remaining MORD techniques and its impact on response detection was one of the main questions for the current work. The simulations showed that when this condition is not satisfied, for all methods except MMSC and MCSM to a large extent, the FP rate obtained is higher than the nominal significance level (5% in Fig. 1). However, for low cross-channel correlations (up to correlation coefficients of about 0.4) the impact is minor. These results demonstrate that MMSC and MCSM are robust to the correlation of background activity in the EEG channels.

Given that there is a clear propensity to inflated FP rates with some of

the MORD techniques, but that methods are robust to low cross-channel correlations, we then addressed the question of whether the methods lead to inflated FP rates in recorded EEG signals. In Fig. 2, the percentage of combinations of channels that passed the first criterion (FP around 5%) reduces with increasing N and no combinations of channels remained within the desired levels of significance for $N \geq 5$ with the exception of MMSC and MCSM – in agreement with the results for the simulated data. This suggests that the noise source activity captured in different channels were sufficiently uncorrelated only when a small number of (optimally selected) channels are included, but as the electrode density increases, the impact of cross-channel correlation becomes strong (as also observed in [23]).

Thus, in the DR analysis only combinations of channels that passed the first criterion (acceptable FP rate) were used. In Fig. 3, the performance of MORD techniques initially increases with increasing N until it reached a maximum, between $N = 2$ and 4, depending on the MORD technique, and then the performance decreased (worsening detection rates and/or no combinations of channels passed the first criterion). The exception was MMSC, which peaked at $N = 7$. These results are in agreement with those reported in some previous works [17,18,36] where it was also observed that additional channels improve detection rates at first, but when including large numbers of additional channels, DR degrade. This is probably due to the extra channels bringing little (independent) new information, but adding more noise to the analysis.

The best combinations of electrodes will involve a compromise between signal to noise ratio (SNR) and correlation between channels. The results indicate that optimal combinations have electrodes placed in the frontal (FC_z, that is, between frontal and central) and parietal regions (Table 1). The electrodes located in the central and frontal region over the scalp have the highest SNR (with the nape as a reference electrode and 80-Hz ASSR, see [37]). However, the performance of some MORD techniques (excluding MMSC and MCSM) tends to decrease if only electrodes from this region are used, probably because of the higher correlation of their EEG signals. The signals from the electrodes of the parietal region have a lower correlation with those more distant in the central and frontal region and a higher SNR than the electrodes in the temporal region. This may explain why they were included in the optimal selection.

When analysing the performance of the MORD techniques in ASSR detection, the aMSC exhibited the best DR performance with an increase of 15.7 % (from a DR of 57.9%–67.0%) in relation to the best univariate detector, though this performance was not significantly different to that of pMSC, aCSM, MMSC, aLFT and pCSM. It may also be noted that, while the FP rates of all methods were not significantly different to the 5%, Table 1 shows that it was below 5% only for MMSC and MCSM (and single channel MSC), and rather higher for the other methods. A larger sample may well have shown that these other methods have a slightly inflated FP rate (and MMSC and MCSM may be significantly conservative). This may further explain why in the current sample aMSC, pMSC and aCSM outperformed MMSC.

It might also be noted that all MORD techniques showed a computational time of less than one second in each of the recordings included in this paper, when using a 4 GHz laptop computer, running Windows 10 and 2016a version of Matlab (the code was also not specifically optimized for computational time). The computational time would therefore not noticeably prolong the ASSR test procedure in clinic or research.

Overall, only the MMSC and MCSM consistently controlled the FP rate with increasing numbers of channels – and consequently the between-channel correlation level. However, some of the other detectors can control the FP rate when a small number of (optimally selected) channels are used. In order to utilize these detectors regardless of combinations of channels, future research could include the correlation level in the estimation of their critical values in order to control the FP rate in the presence of correlated background noise. In the absence of

such corrections, the current results suggest that only MMSC and MCSM are consistently robust, even if not necessarily the most sensitive methods.

The use of MORD methods can clearly increase the detection of ASSR signals. However, this does come at the cost, in set-up time and equipment, of a multichannel recording. Unless multiple EEG channels are already being used (e.g. in studies of source localization), the benefit of a shorter stimulation time needed to detect responses (or increased DR) must be balanced against the longer time required to set up the recording.

6. Conclusion

In this work, the performances of nine MORD techniques were evaluated in the detection of 80 Hz ASSRs. The results showed that detection performance can be increased when more than one signal is used together with an appropriate MORD method. Up to four channels were found to be beneficial with most MORD methods, and MMSC showed peak performance with 7 channels. Beyond this range, detection rates drop. Caution is required with many (simpler) MORD methods, as FP rates can be inflated due to cross-channel correlation. Under the null hypothesis of lack of evoked responses, the MMSC technique is immune to such correlation and should therefore be recommended as the most robust method. In general, the combination of electrodes for the best performances of MORD techniques was obtained by combining electrodes of the frontal and parietal regions.

A comparison of the performance of MORD techniques showed that the new techniques proposed in this work, aMSC followed by pMSC and aCSM, exhibited the highest performances in ASSR detection (though not significantly higher than MMSC, aLFT and pCSM). The best case result - aMSC with $N = 3$ - achieved a 15.7 % increase in detection rate when compared to the best univariate case, confirming that multichannel EEG with MORD detectors are a promising tool for ASSR detection.

In summary, the main advantage of the new proposed techniques over the conventional protocols that encompass both univariate and multivariate detectors is that a superior performance may be obtained with ASSR data. However, they require attention in the number of signals selection, since correlation between the channels may inflate the detection rate at the expense of increasing the false positive rate. The superior performance observed in the present work was not significantly higher than the MMSC, for which false positive rate remain unaltered in spite of the correlation between the electrodes.

CRediT authorship contribution statement

Tiago Zanotelli: Conceptualization, Methodology, Validation, Formal analysis, Writing - original draft. **Quenaz Bezerra Soares:** Methodology, Visualization, Writing - original draft. **David Martin Simpson:** Methodology, Writing - review & editing. **Antonio Mauricio Ferreira Leite Miranda de Sá:** Writing - review & editing. **Eduardo Mazoni Andrade Marçal Mendes:** Methodology, Validation, Writing - review & editing. **Leonardo Bonato Felix:** Investigation, Methodology, Writing - review & editing.

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Declaration of Competing Interest

The authors report no declarations of interest.

Appendix A

A.1 Proof that cross-correlation between the channels does not affect the distribution of the MMSC under H_0

The MMSC for N correlated channels is shown below. Let $\mathbf{y} = [y_1[k], y_2[k], \dots, y_N[k]]^T$ (with “T” superscript denoting transpose) be the vector of the N discrete-time uncorrelated channels ($E[y_p \cdot y_q] = 0, \forall p \neq q$), assuming that $\mathbf{z} = [z_1[k], z_2[k], \dots, z_N[k]]^T$ is given by

$$\mathbf{z} = \mathbf{A} \cdot \mathbf{y}, \quad (\text{A.1})$$

where $\mathbf{A}$ is a ($N \times N$) mixture matrix. As the DTF is a linear operator, it follows that $\mathbf{Z}(f_0) = \mathbf{A} \cdot \mathbf{Y}(f_0)$, where $\mathbf{Z}(f_0)$ and $\mathbf{Y}(f_0)$ are the Fourier Transform at frequency f_0 of $\mathbf{z}$ and $\mathbf{y}$, respectively. The MMSC of $\mathbf{z}$ is obtained from Eq. 11:

$$MMSC_z(f_0) = \frac{\mathbf{V}_z^H(f_0) \hat{\mathbf{S}}_{zz}^{-1}(f_0) \mathbf{V}_z(f_0)}{M}, \quad (\text{A.2})$$

Substituting $\hat{\mathbf{S}}_{zz}^{-1}(f_0) = (\mathbf{A}^H)^{-1} \hat{\mathbf{S}}_{yy}^{-1}(f_0) \mathbf{A}^{-1}$ and $\mathbf{V}_z(f_0) = \mathbf{A} \cdot \mathbf{V}_y(f_0)$, in expression (A.2) we have:

$$\begin{aligned} MMSC_z(f_0) &= \frac{\mathbf{V}_y^H(f_0) \mathbf{A}^H (\mathbf{A}^H)^{-1} \hat{\mathbf{S}}_{yy}^{-1}(f_0) \mathbf{A}^{-1} \mathbf{A} \mathbf{V}_y(f_0)}{M} \\ &= \frac{\mathbf{V}_y^H(f_0) \hat{\mathbf{S}}_{yy}^{-1}(f_0) \mathbf{V}_y(f_0)}{M}. \end{aligned} \quad (\text{A.3})$$

The MMSC of $\mathbf{z}$ and $\mathbf{y}$ are thus equal, and the correlation between the channels does not affect the distribution under H_0 . The critical values obtained from uncorrelated channels can thus also be applied when channels are correlated, without an inflation of the FP rate.

A.2 Proof that cross-correlation between the channels does not affect the distribution of the MCSM under H_0

Let N uncorrelated channels divided into M non-overlapping windows, in the condition of H_0 (absence of response). The phase of the i -th window and j -th channel ($\theta_{ij}(f_0)$) is a circular variable with uniform distribution between $0-2\pi$. The average phase of the i -th window $\bar{\theta}_i(f_0)$ (i.e. the mean direction of $\theta_{i1}(f_0) + \theta_{i2}(f_0) + \dots + \theta_{iN}(f_0)$) it is also a circular variable with uniform distribution between $0-2\pi$ [38].

Now, considering a mixture between the phases of the channels represented by the $\mathbf{a} = [a_1, a_2, \dots, a_N]$, the average mixed phase ($\bar{\theta}_i^M(f_0)$) is given by mean direction of $a_1 \theta_{i1}(f_0) + a_2 \theta_{i2}(f_0) + \dots + a_N \theta_{iN}(f_0)$. According to [34, pg 65–66], the uniform distribution is invariant under rotation and reflection in the circle, so $a_j \theta_{ij}(f_0)$ also has uniform distribution between $0-2\pi$. Thus $\bar{\theta}_i(f_0)$ and $\bar{\theta}_i^M(f_0)$ (and consequently $\theta_{ij}(f_0)$), although they may have different values, have the same distribution, i.e. uniform distribution between $0-2\pi$.

As the correlation between the channels does not affected the independence between the windows (it depends on whether the windows are overlapping or not), it can be concluded that the distribution of the MCSM under H_0 is not affected by the correlation between the channels.

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